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Heat Generation by Chemical Reaction Underground to Assist Steam towards Heavy Oil Recovery Enhancement

Zhiqian Yan, National Key Laboratory of Petroleum Resources and Engineering, China University of Petroleum, China / College of Petroleum Engineering, China University of Petroleum, China; Yiqiang Li, National Key Laboratory of Petroleum Resources and Engineering, China University of Petroleum, China / College of Petroleum Engineering, China University of Petroleum, China; Wenjie Wang, National Key Laboratory of Petroleum Resources and Engineering, China University of Petroleum, China / College of Petroleum Engineering, China University of Petroleum, China; Maojie Chai, National Key Laboratory of Petroleum Resources and Engineering, China University of Petroleum, China / College of Petroleum Engineering, China University of Petroleum, China; Jianqiang Zi, National Key Laboratory of Petroleum Resources and Engineering, China University of Petroleum, China / College of Petroleum Engineering, China University of Petroleum, China; Zheyu Liu, National Key Laboratory of Petroleum Resources and Engineering, China University of Petroleum, China / College of Petroleum Engineering, China University of Petroleum, China; Tao Song, National Key Laboratory of Petroleum Resources and Engineering, China University of Petroleum, China / College of Petroleum Engineering, China University of Petroleum, China

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Abstract

Heavy oil accounts for more than 70% of crude oil resources worldwide. Considered as main heavy oil recovery technology, steam injection has problems such as difficulties in improving underground steam temperature, dryness and high energy consumption with large carbon emission of steam preparation on the ground. How to improve heat utilization efficiency is critical issue for heavy oil thermal recovery under the background of carbon neutrality. Herein, we propose to produce a chemical reaction under the continuous supply of reactants and steam, which can generate hydrocarbon and release heat step by step towards high steam quality injection. Conduct chemical reaction assisted steam physics simulation experiments at the core scale to explore the influence of chemical reactions on the viscosity and heavy oil recovery. Conduct numerical simulation research on thermal hydrocarbon synergistic steam, determine the optimal conditions for chemical reactions, and clarify the mechanism of thermal hydrocarbon synergistic steam to enhance heavy oil recovery. It was found that the product of the reaction could decrease the viscosity of heavy oil nearly 50% even at the relatively low temperature. Compare to the steam flooding, chemical reaction assist steam could further increase oil recovery factor more than 23% and significantly improve the quality of steam. Through numerical simulation, the optimal steam injection temperature for chemical reaction assisted steam is determined to be 250°C, and the hydrocarbon concentration 3 mol%. This method can solve the

problem of steam injection for heavy oil production in the deeply-buried reservoir, which will provide basic theory and method guidance for the breakthrough of disruptive technology of large-scale green, low-carbon and efficient development of heavy oil.

Keyword: Heavy Oil, Heat Generation, Hydrocarbon, Chemical Reaction, Enhance Oil Recovery

Introduction

With the continuous depletion of conventional oil and gas resources and the sustained growth of energy demand, countries around the world have begun to focus on the recovery and utilization of unconventional oil and gas resources such as heavy oil^[1-3]. Due to the high viscosity and low mobility of heavy oil, there are certain technical difficulties in its efficient development^[4-5]. Currently, there are mainly heavy oil thermal recovery technologies such as steam huff and puff, steam flooding, and steam assisted gravity drainage (SAGD)^[6-8]. However, the above-mentioned single technologies all have technical bottlenecks, making it difficult to effectively develop newly added or remaining heavy oil resources with low reserve grades, poor crude oil properties, and complex geological conditions. The low utilization rate of steam heat and large heat loss during the development process, as well as the cost of steam generation and the emission of CO₂ and other gases, affect the recovery of heavy oil^[9]. New technologies are needed to achieve efficient development of heavy oil resources^[10-12].

The existing technologies for in-situ catalytic modification of heavy oil, gas injection assisted steam flooding, solvent assisted steam flooding, and electric heating assisted development of heavy oil have significantly contributed to the evolution of heavy oil recovery technologies from the aspects of improving crude oil properties, enhancing thermal efficiency, and environmental emissions. Yusuf A et al.^[13] used a catalytic system consisting of nickel oleate catalyst and hydrogen donor glycerol to catalyze the upgrading of Oman heavy oil. The results showed that the catalytic system could effectively reduce the viscosity of heavy oil and convert heavy components into light components. Rafiee et al.^[14] used numerical simulation techniques to study the mechanism of electromagnetic heating in SAGD process. The research results showed that the temperature distribution generated by electromagnetic heating was more uniform than that in SAGD, and the energy utilization rate was higher. After calculation, 92% of the energy was transferred to the reservoir. In recent years, multi-component thermal fluid has gradually gained attention as a new injection medium. The main mining mechanisms of multi-component thermal fluid include reducing the viscosity of crude oil, thermal expansion, improving oil phase permeability, gravity differentiation, enhancing rock permeability, and utilizing the Jamin effect^[15]. Li et al.^[16] studied the effect of CO₂ on improving displacement efficiency by conducting displacement experiments on heavy oil in Mao 8 block. The displacement experiments showed that the recovery degree of CO₂ and viscosity reducer assisted steam flooding was better than that of viscosity reducer assisted steam flooding, and the recovery degree increased by 11.4%, indicating that CO₂ has a significant effect on improving recovery degree in the thermochemical composite system. W. Y. Svrcak et al.^[17] measured the viscosity of the Athabasca crude oil system with N₂, CO₂, CH₄. The experimental results show that CO₂ has the highest solubility and the best viscosity reduction effect, followed by CH₄, and N₂ has the lowest solubility and the worst viscosity reduction effect. Zare et al.^[18] conducted a study on the mechanism of hydrocarbon solvent-CO₂ synergistic assisted SAGD to enhance oil recovery through two-dimensional physical simulation experiments. The results indicate that the high solubility of C₆ in heavy oil, the diffusion of CO₂ in heavy oil, and the reduction of heat loss in the steam chamber are the main mechanisms for improving the oil recovery of this development method. There is relatively little research on the use of chemical reactions to generate hydrocarbons and release heat to assist steam flooding, and the mechanism for improving oil recovery is not yet clear. Further research is needed.

This article proposes a method of steam assisted heavy oil recovery using heat and hydrocarbons generated by underground chemical reactions. Based on the hydrocarbons generated by the Fischer Tropsch reaction and the heat released by the reaction, a multi-component thermal fluid is formed to achieve efficient development of heavy oil. This study measured the viscosity of heavy oil dissolved in different concentrations of hydrocarbons, and elucidated the effect of thermal hydrocarbon synergy on the viscosity of heavy oil. In addition, chemical reaction assisted steam core displacement experiments and numerical simulations were conducted to further elucidate the mechanism by which the heat generated by underground chemical reactions and hydrocarbons synergistically enhance heavy oil recovery.

Experiments and methods

Experimental materials

The crude oil used in the experiment is heavy oil from Karamay oilfield with a viscosity of 5462 mPa·s at 25°C. The gas used in the experiment is hydrocarbon with a purity of 99.9%. The water used in the experiment is simulated formation water configured. All chemicals are of analytical grade for direct use without further purification. The cores used in the experiment are natural cores taken from the oil field. The physical properties of natural cores are shown in Table 1.

Table 1—Parameters of cores in core flooding experiments

Number	Length(mm)	Diameter(mm)	Permeability(mD)	Porosity(%)	Oil saturation(%)
①	30.21	2.49	807.14	31.92	75.41
②	30.02	2.50	812.45	32.02	76.05
③	30.08	2.48	804.68	31.02	76.12

Experimental instruments

The main experimental instruments include high temperature and high pressure PVT reactor, falling ball viscometer, steam generator, ISCO pump, high-temperature oven, core holder, piston container, gas cylinder.

Experimental study on the viscosity reduction of heavy oil by hydrocarbon

Fill the PVT cylinder of the high-temperature and high-pressure PVT reactor with crude oil and hydrocarbon in a certain proportion, and continuously rotate to mix them evenly. After sufficient mixing, inject the oil and gas mixture into the high-temperature and high-pressure falling ball viscometer to measure its viscosity. Change the hydrocarbon concentration and temperature inside the PVT cylinder, and measure the viscosity of heavy oil at different hydrocarbon contents and temperatures according to the above steps.

Thermal-hydrocarbon assisted steam core flooding experiments

This study designed three sets of experiments, the first set being steam flooding, and the second and third sets being thermal-hydrocarbon assisted steam flooding. Exploring the Effect of heat generation by chemical reaction underground to assist steam on heavy Oil development by comparing the recovery of two experiments. This section of the experiment involves the following specific steps: (1) Use a vacuum pump to evacuate the cores and then saturate it with simulated formation water; (2) Weigh the wet weight of the core and calculate the pore volume of the core based on the difference between the wet weight and dry weight; (3) Inject heavy oil into the core at a rate of 0.1mL/min at 80 °C until there is no more water production in the outlet section, and calculate the oil saturation of the core; (4) Connect the experimental process and start flooding according to the experimental scheme, recording the pressure, oil production, and water production at different times. The experimental process is shown in Fig.1. The experimental parameters are shown in Table 2 (5) Change the experimental parameters and repeat steps (2)-(4).

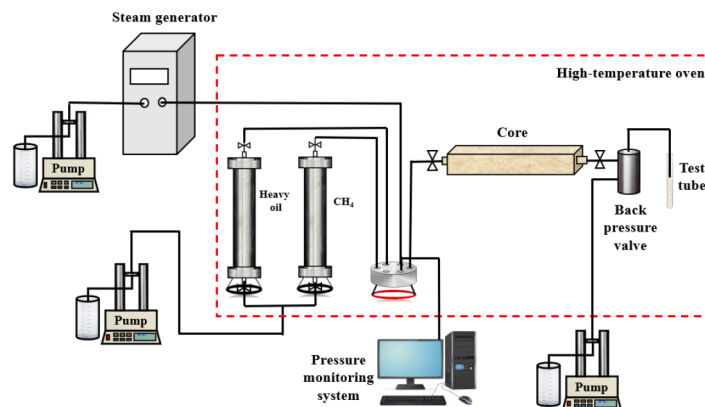


Figure 1—core flooding experimental setup

Table 2—Core flooding experiments scheme

Number	Experimental temperature (°C)	Steam injection rate (mL/min)	Displacing medium	CH ₄ concentration (mol%)
①	200	3	Steam	/
②	200	3	Steam+CH ₄	10
③	200	3	Steam+CH ₄	20

Numerical simulation of thermal-hydrocarbon assisted steam flooding

Due to the limitations of physical simulation experiments, numerical simulation studies were conducted on heat generating steam generated by chemical reactions. The steam injection temperature and hydrocarbon concentration were optimized to determine the experimental parameters that can achieve the best development effect. By comparing the distribution of temperature field, pressure field, and oil saturation field, the mechanism of thermal-hydrocarbon assisted steam enhanced oil recovery is clarified.

A homogeneous model is established based on the physical properties and permeability characteristics of natural rock cores. The model consists of 9000 grid nodes ($30 \times 30 \times 10$), as shown in Fig.2. The settings of other geological parameters are shown in Table 3. Set up one production well and one injection well on both sides respectively. The fluid parameters are set based on the viscosity temperature curve of heavy oil and the results of phase permeability experiments. The specific parameter settings are shown in Fig.3 and Fig.4.

Table 3—Other physical parameters of geological model

Parameter	Values
Porosity / %	27
Permeability / mD	1000
Initial oil saturation / %	65
Initial temperature of reservoir / °C	20
Initial pressure of reservoir / MPa	4.3
Solution gas-oil ratio	8
Rock volumetric heat capacity / ($\text{kJ} \cdot \text{m}^{-3} \cdot ^\circ\text{C}^{-1}$)	1500
Thermal conductivity of rocks / ($\text{kJ} \cdot \text{m}^{-1} \cdot \text{d}^{-1} \cdot ^\circ\text{C}^{-1}$)	915
Thermal conductivity of oil / ($\text{kJ} \cdot \text{m}^{-1} \cdot \text{d}^{-1} \cdot ^\circ\text{C}^{-1}$)	31
Thermal conductivity of formation water / ($\text{kJ} \cdot \text{m}^{-1} \cdot \text{d}^{-1} \cdot ^\circ\text{C}^{-1}$)	52
Thermal conductivity of formation steam / ($\text{kJ} \cdot \text{m}^{-1} \cdot \text{d}^{-1} \cdot ^\circ\text{C}^{-1}$)	0.14
Steam injection speed / $\text{m}^3 \cdot \text{d}^{-1}$	100

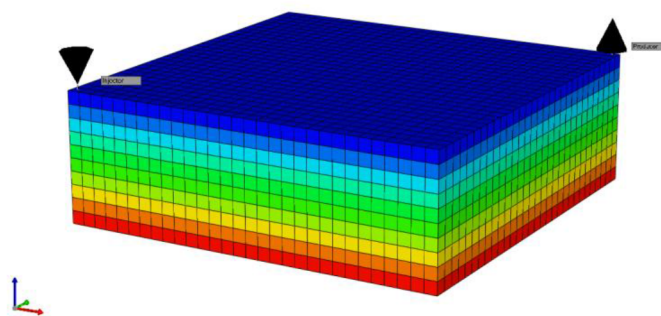


Figure 2—Numerical simulation model diagram

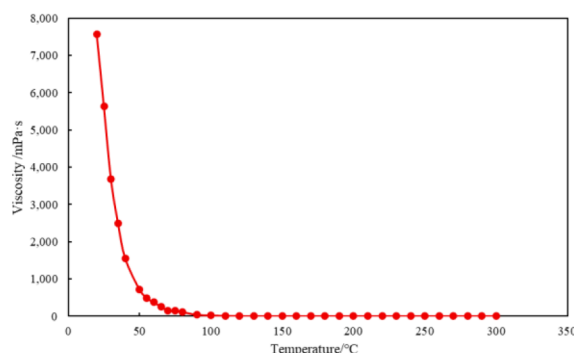


Figure 3—Crude oil viscosity temperature curve

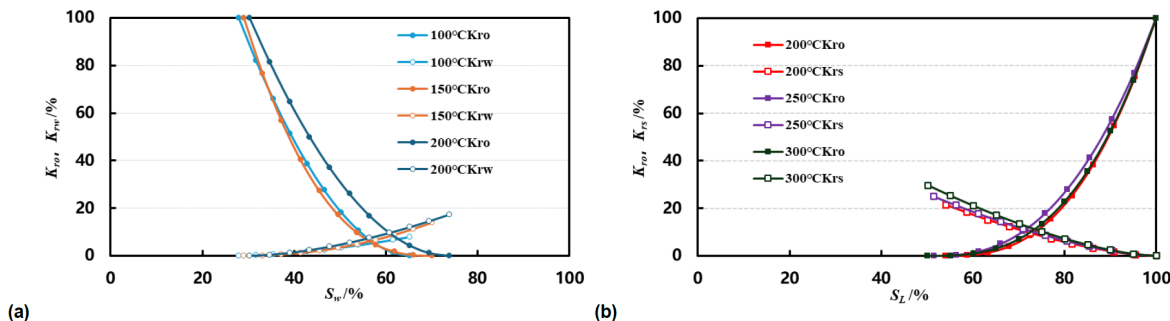


Figure 4—(a) Oil-Water two phase relative permeability curve (b) Oil-Steam two phase relative permeability curve

Results and discussion

The viscosity reduction effect of hydrocarbon on heavy oil

The viscosity reduction effect of heavy oil after dissolving different concentrations of CH_4 is shown in Fig.5(a). The original viscosity of the heavy oil used in the experiment is 791 mPa·s at 50 °C, and 5-25mol % CH_4 is dissolved into the heavy oil in the PVT reactor at this temperature. After sufficient dissolution, the viscosity of crude oil dissolved in 5mol% CH_4 decreased to 543 mPa·s, with a viscosity reduction rate of 31.35%. When the dissolved CH_4 concentrations are 10mol%, 15mol%, 20mol%, and 25mol%, the viscosity of crude oil decreases to 405 mPa·s, 397 mPa·s, 388 mPa·s, and 382 mPa·s. When the CH_4 concentration increases from 0 to 10mol%, the viscosity of heavy oil decreases significantly. Subsequently, as the concentration of CH_4 increases, the decrease in viscosity of heavy oil becomes smaller and smaller. After the dissolution of CH_4 in heavy oil, the intermolecular forces between liquid/liquid molecules in the system are transformed into intermolecular forces between gas/liquid molecules, greatly reducing the frictional force. The layered macromolecular structure of the original system's gelatinous asphaltene

stacking is also destroyed, resulting in a significant decrease in viscosity. But as the concentration of CH_4 increases, the solubility of CH_4 in heavy oil reaches its upper limit, and any further increase in CH_4 cannot dissolve into the heavy oil. Therefore, the increase in viscosity reduction rate for heavy oil gradually decreases. Therefore, the solubility of CH_4 in crude oil determines its viscosity reduction effect.

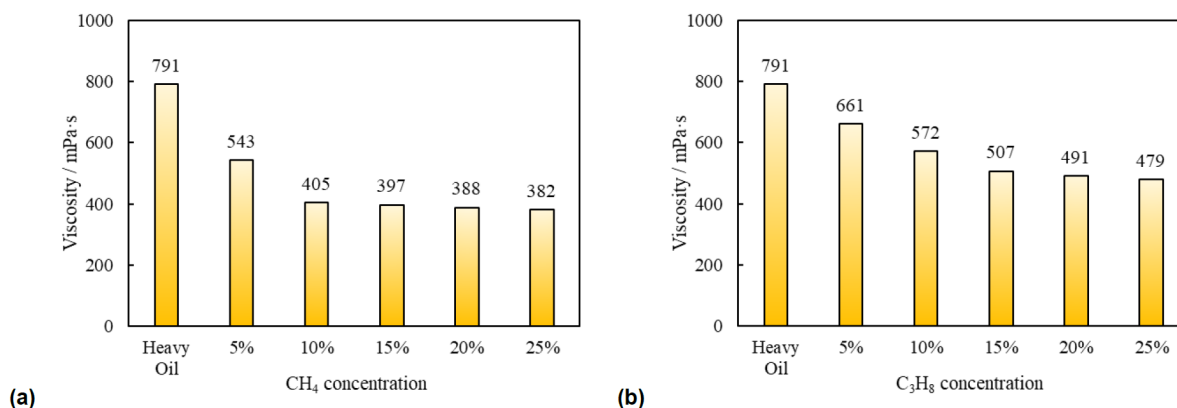


Figure 5—(a) The viscosity reduction effect of CH_4 on heavy oil (b) The viscosity reduction effect of C_3H_8 on heavy oil

The viscosity reduction effect of heavy oil after dissolving different concentrations of C_3H_8 is shown in Fig.5(b). After sufficient dissolution, the viscosity of crude oil dissolved in 5mol% C_3H_8 decreased to 661 mPa·s. When the dissolved C_3H_8 concentrations are 10mol%, 15mol%, 20mol%, and 25mol%, the viscosity of crude oil decreases to 572 mPa·s, 507 mPa·s, 491 mPa·s, and 479 mPa·s. When the concentration of C_3H_8 is less than 15mol%, the viscosity reduction effect on heavy oil is significant with the increase of C_3H_8 concentration. When the concentration of C_3H_8 is greater than 15%, the increase in viscosity reduction rate for heavy oil gradually decreases. Overall, the viscosity reduction effect of C_3H_8 on heavy oil is inferior to that of CH_4 . Therefore, CH_4 was chosen as the hydrocarbon for laboratory use in subsequent experiments.

Thermal-hydrocarbon assisted steam core flooding experiments

Through static viscosity reduction experiments, it has been proven that CH_4 has a better viscosity reduction effect on heavy oil. Based on the above experimental results, CH_4 was selected as the hydrocarbon to conduct core flooding experiments, exploring the influence of thermal hydrocarbons on improving the recovery of heavy oil in porous media of reservoirs.

The recovery curve of steam flooding experiment is shown in Fig.6. At the beginning of steam flooding, the water content and pressure rapidly increase, and a large amount of heavy oil in the core is driven out, resulting in a rapid increase in recovery. As the injection amount increases, the water cut fluctuates between 80-95%, and the pressure begins to decrease. As the amount of steam injection increases, crossover channels begin to appear. At this point, subsequent injected steam preferentially advances along the crossover channels, and the expansion speed of the affected area slows down significantly, resulting in a gradual flattening of the increase in recovery. When the injection volume reaches 2.0 PV, the moisture content remains stable at around 97%, and there is no significant change in recovery at this point. When the injection volume reached 2.7PV, steam flooding reached its limit and the final recovery was 46.94%.

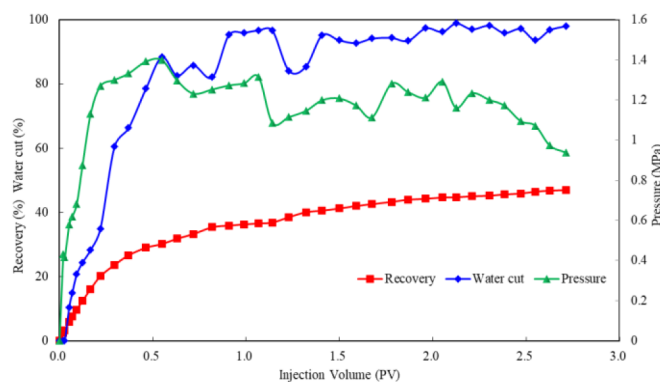
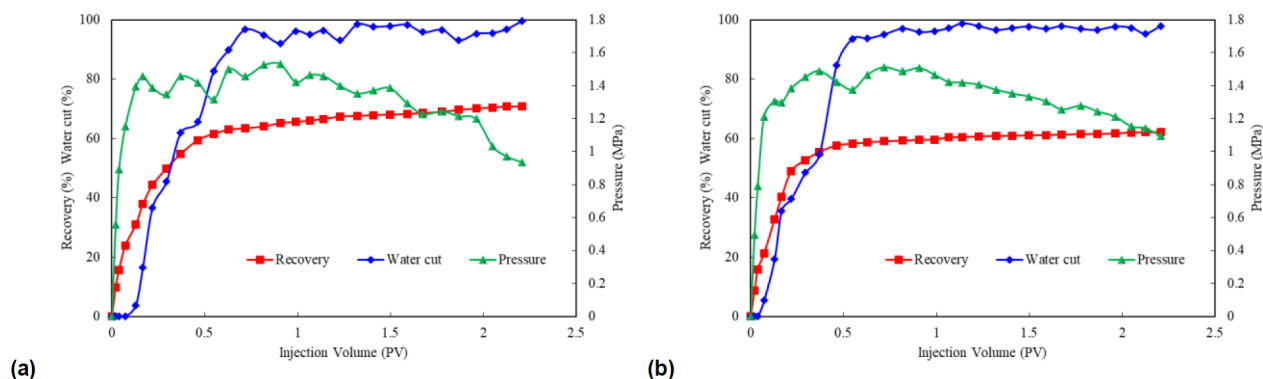


Figure 6—Steam flooding

The recovery curve of 10mol% CH₄ assisted steam flooding is shown in Fig.7(a). When the CH₄ injection concentration is 10mol%, the pressure rapidly increases in the early stage of displacement, and the recovery of heavy oil increases significantly, compared to the recovery of simple steam flooding. After the injection volume reached 0.55PV, the increase in recovery gradually slowed down, and the water cut remained stable at around 95%. In the early stage, due to the injection of CH₄, it can dissolve in heavy oil and significantly reduce its viscosity, making it easier to use and greatly improving the recovery. Moreover, CH₄ can reach the remaining oil areas that have not been affected by the steam flooding process, reducing viscosity and displacing the remaining oil that is difficult to displace, thereby improving the recovery. When the injection volume reached 2.2PV, thermal-hydrocarbon assisted steam flooding reached its limit and the final recovery was 70.79%. The recovery curve of 20mol% CH₄ assisted steam flooding is shown in Fig.7(b). In the early stage of displacement, the overall trend of each curve is basically the same as the curve when the CH₄ concentration is 10mol%, with a rapid increase in water cut and a significant increase in recovery. When the injection volume reaches 0.44PV, the increase in recovery gradually decreases. Afterwards, the degree of recovery did not change significantly, the water cut remained stable at around 96%, and the pressure gradually decreased. When the displacement reaches its limit, the final recovery is 62.43%, which is 8.36% lower than the recovery when the CH₄ concentration is 10mol%. The reason is that the increase in the amount of CH₄ injected makes CH₄ more prone to gas channeling, which makes it difficult to fully mix with heavy oil to reduce viscosity, resulting in a decrease in recovery.

Figure 7—(a) 10mol% CH₄ assisted steam flooding (b) 20mol% CH₄ assisted steam flooding

Optimization of thermal-hydrocarbon assisted steam parameters

Based on the established numerical model, carry out parameter optimization of thermal-hydrocarbon assisted steam flooding. Simulate and optimize the steam injection temperature and CH₄ concentration,

clarify the mechanism of thermal hydrocarbon assisted steam flooding to improve oil recovery, and obtain the optimal parameter.

The oil saturation field of steam flooding is shown in Fig.8 (a), and the oil saturation field of CH₄ assisted steam flooding is shown in Fig.8 (b). The steam propulsion effect of steam flooding is poor. The utilization range of heavy oil is relatively small, mostly concentrated near injection wells, and the remaining oil content is about 50%. The oil saturation of CH₄ assisted steam flooding is lower than that of steam flooding, and the utilization range of heavy oil is larger. The remaining oil content near the injection well is significantly lower than that of steam flooding, which has a better effect on the development of heavy oil. The temperature field of steam flooding is shown in Fig.9 (a), and the temperature field of CH₄ assisted steam flooding is shown in Fig.9 (b). The steam chamber expansion range of steam flooding is relatively small, and the temperature of the steam chamber is low, with obvious cold and hot distribution. The heat loss of steam is relatively large, which cannot significantly increase the temperature of the reservoir and is not conducive to the development of heavy oil. CH₄ assisted steam flooding generates heat through chemical reactions, which increases the temperature inside the steam chamber, expands the range of the steam chamber, and creates a clear thermal front, making it more conducive to the development of heavy oil. The synergistic effect of CH₄ significantly enhances the thermal leading edge propulsion capability and oil and gas drive efficiency.

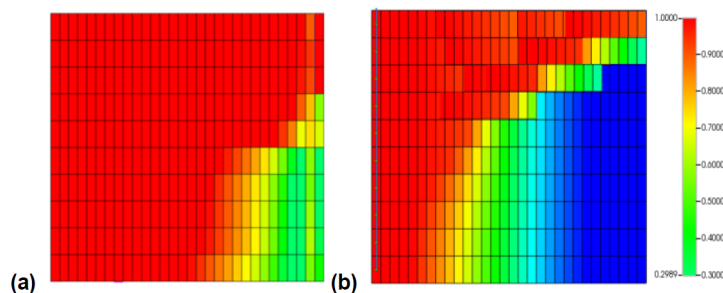


Figure 8—(a) The oil saturation field of steam flooding (b) The oil saturation field of CH₄ assisted steam flooding

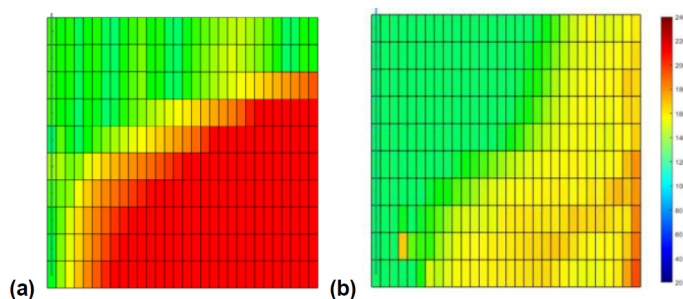


Figure 9—(a) The temperature field of steam flooding (b) The temperature field of CH₄ assisted steam flooding

Based on the established model, four CH₄ injection concentrations and four steam injection temperatures were set for parameter optimization. The recovery curves under different injection parameters are shown in Fig.10. When the concentration of CH₄ is 3mol%, the recovery in the early stage is relatively low. With the increase of development time, the recovery in the later stage grows rapidly, and the final recovery is the highest, resulting in the best development effect. The results indicate that an increase in CH₄ concentration does not promote an increase in recovery, and the optimal injection concentration of CH₄ is 3mol%. When the injection temperature of steam is 250°C, the final recovery is the highest, and the development effect of heavy oil is the best. The optimal steam injection temperature is 250°C.

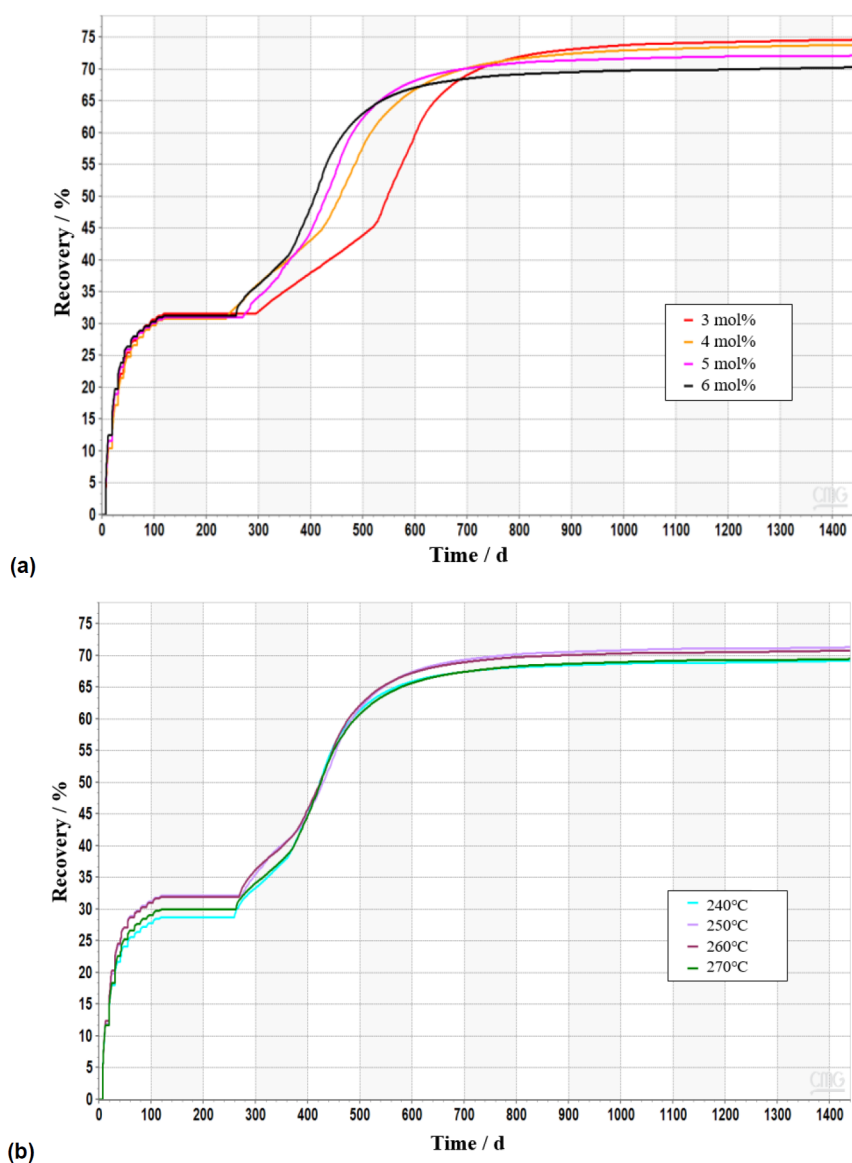


Figure 10—(a) Recovery of heavy oil at different CH_4 concentrations
(b) Recovery of heavy oil at different steam injection temperatures

Conclusions

1. The hydrocarbons generated by chemical reactions can dissolve in heavy oil, significantly reducing its viscosity. CH_4 has the best viscosity reduction effect on heavy oil, achieving a viscosity reduction rate of up to 51.7%.
2. Physical simulation experiments have demonstrated that it has been proven that chemical reaction assisted steam flooding can effectively improve the recovery of heavy oil. When the concentration of CH_4 produced by the reaction is 10mol%, the recovery of heavy oil can reach 70.79%, which is 23.85% higher than that of single steam flooding.
3. Numerical simulations have shown that chemical reaction assisted steam flooding has a better development effect on heavy oil. The heat generated by chemical reactions can expand the range of heat waves, improve the efficiency of heat utilization, and make the temperature distribution of the reservoir more uniform; The CH_4 generated by the reaction can effectively synergize with steam to improve the recovery of heavy oil. The optimal steam injection temperature for chemical reaction assisted steam is determined to be 250°C, and the hydrocarbon concentration range is 3 mol%.

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